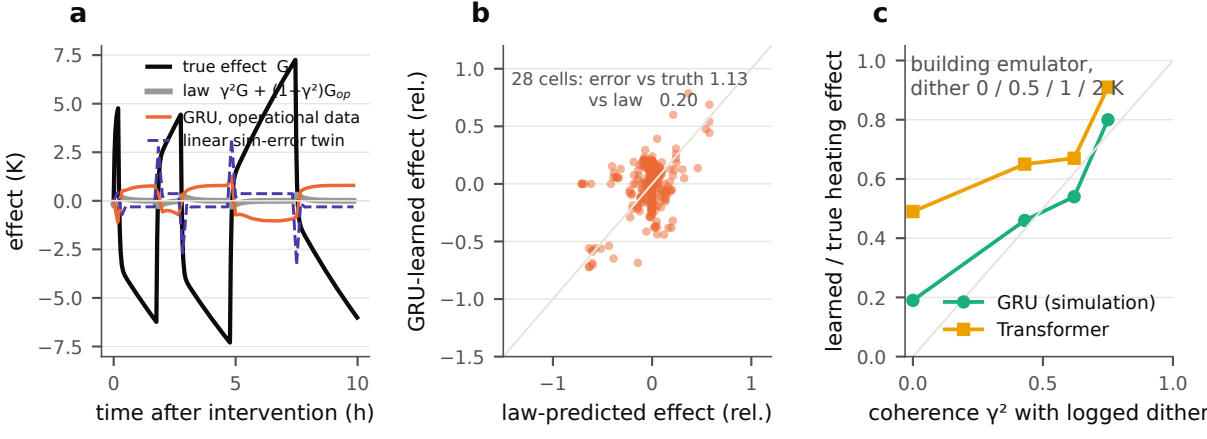
a
b
c
true effect G
law γ²G + (1−γ²)G_op
GRU, operational data
linear sim-error twin
effect (K)
time after intervention (h)
28 cells: error vs truth 1.13
vs law 0.20
GRU-learned effect (rel.)
law-predicted effect (rel.)
building emulator,
dither 0 / 0.5 / 1 / 2 K
GRU (simulation)
Transformer
learned / true heating effect
coherence γ² with logged dither